

major war. But the Middle East offered plenty of scope for US power-projection in which naval forces could play a central role.

SUPPORTING LAND FORCES

The largest naval operations since 1945 have been conducted in support of land forces, with firepower deployed for the bombardment of targets on land. During the Korean War and the Vietnam War, big naval guns and carrier aircraft were used by the Americans and their allies on a massive scale to provide fire support and to devastate targets in enemy territory. This pattern was repeated in Iraq and Afghanistan in the 1990s and more recently—although with ship-launched missiles replacing the big warships' heavy guns.

These large-scale operations were largely uncontested at sea, but during the same period several of the world's smaller navies engaged in some sharp, if mostly limited, naval combat that provided the opportunity for real-life use of developing new technologies and tactics. The Middle East and the Indian subcontinent were the scenes of interesting encounters, including the successful use of a ship-to-ship missile in the sinking of the Israeli destroyer *Eilat* by Egyptian missile boats in 1967. The Falklands War of 1982 was an unequal contest between the Royal Navy and Argentina's small and ageing fleet, but a more balanced contest between the British ships and shore-based Argentinian aircraft armed with advanced missiles as well as bombs.

AWESOME POWER

By the early 21st century the firepower of navies was far beyond anything that had previously existed, even if nuclear warheads were left out of account. The aircraft and missiles of the fleets assembled by the United States and its allies for operations in Afghanistan and Iraq constituted an awesome concentration of force and sophisticated technology. Nuclear submarines were among the technological wonders of the modern world. Whether such supremely impressive naval forces were a weapon that would continue to suit 21st-century strategic needs in the age of the "War on Terror" remained to be demonstrated.



Quiet sentinels
Modern nuclear-powered submarines such as HMS *Astute*, launched by the Royal Navy in 2007, are able to remain submerged for months at a time.

WEAPONS AND TECHNOLOGY

BALLISTIC MISSILES

Unlike cruise missiles, which take a guided flight path to their targets, ballistic missiles are guided only during the relatively brief powered multi-stage lift-off and initial flight, which takes the missile to sub-orbital altitudes. Thereafter the missile flies on using its own ballistic, free-flight energy, eventually deploying nuclear or conventional warheads at supersonic speeds to the target area. Ranges of these missiles vary from a few hundred miles to more than 3,418 miles (5,500 km) in inter-continental models. Naval ballistic missiles are primarily submarine-launched weapons. The first operational naval model was the Polaris, which entered service with the US Navy in 1961 after a five-year development program. The latest Trident ballistic missiles have a range of 7,000 miles (11,300 km), reach speeds of 18,000 mph (29,000 kph) in sub-orbit, and use star-sighting and inertial guidance systems.



Polaris nuclear missile
The Polaris missile was a submarine-launched, solid-fuel, nuclear-armed, ballistic missile (SLBM), which was carried by British submarines from 1968. It was later replaced by Trident I.